Name: _____

BME 364
MECHANISMS OF ALLERGIC DISEASE
Lab Manual

Group I

MTT cell viability assay

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MTT cell viability assay

Objective: To evaluate cell viability and proliferation by quantifying mitochondrial-dependent reduction of MTT (3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyl tetrazolium bromide) into formazan crystals.



3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide **(MTT)**

(*E*,*Z*)-5-(4,5-dimethylthiazol-2-yl)-1,3-diphenylformazan **(Formazan)**

Materials

- 12 mM MTT (5 mg/ml)
- Culture medium DMEM
- Dimethyl sulfoxide (DMSO)
- Sterile 96-well tissue culture plates
- Pipettes and sterile tips
- CO_2 incubator (37°C, humidified)
- Microplate reader capable of measuring absorbance at 540 nm

Solutions

P80 Treatment Solutions

For 10 mL of 1% (v/v) P80 stock solution:

1. In a sterile laminar flow hood, add 9.9 mL of sterile complete culture medium (warmed to 37°C) to a 15 mL sterile conical tube.
2. Add 100 µL of sterile P80 to the medium. (Calculation: 1% of 10 mL = 0.1 mL = 100 µL).
3. Vortex vigorously until the P80 is completely dissolved.

Serial Dilution of P80:

1. Label 7 sterile 2 mL microcentrifuge tubes: “1%”, “0.5%”, “0.25%”, “0.125%”, “0.0625%”, “0.03125%” and “VC”.
2. Add 1.6 mL of your 1% P80 Treatment Stock to the “1%” tube.
3. Add 0.8 mL of complete culture medium to the five dilution tubes (“0.5%” to “0.03125%”) and 1.6 mL to the “VC” tube (12 control wells).
4. Perform a 2-fold serial dilution:
5. Transfer 0.8 mL from the “1%” tube to the “0.5%” tube. Vortex to mix.
6. Transfer 0.8 mL from the “0.5%” tube to the “0.25%” tube. Vortex to mix.
7. Repeat transfer and vortex steps down to the “0.03125%” tube (1:32 of the 1% stock).
8. The "VC" tube contains only complete culture medium (0% P80).

Plate map

This plate map is specifically designed for your P80 dose-response experiment.

	1	2	3	4	5	6	7	8	9	10	11	12
A	PBS	PBS	PBS	PBS	PBS	PBS	PBS	PBS	PBS	PBS	PBS	PBS
B	PBS	VC	1%	0.50%	0.25%	0.13%	0.06%	0.03%	VC	BLK	PBS	PBS
C	PBS	VC	1%	0.50%	0.25%	0.13%	0.06%	0.03%	VC	BLK	PBS	PBS
D	PBS	VC	1%	0.50%	0.25%	0.13%	0.06%	0.03%	VC	BLK	PBS	PBS
E	PBS	VC	1%	0.50%	0.25%	0.13%	0.06%	0.03%	VC	BLK	PBS	PBS
F	PBS	VC	1%	0.50%	0.25%	0.13%	0.06%	0.03%	VC	BLK	PBS	PBS
G	PBS	VC	1%	0.50%	0.25%	0.13%	0.06%	0.03%	VC	BLK	PBS	PBS
H	PBS	PBS	PBS	PBS	PBS	PBS	PBS	PBS	PBS	PBS	PBS	PBS

PBS: Edge wells filled with sterile PBS.
BLK (Blank): Media + MTT + DMSO (No cells).
VC (Vehicle Control): Cells + Media (0% P80) + MTT + DMSO, in columns 2 and 9 on both sides of the dose series. This is your "Control" group and 100% viability reference (mean of all 12 wells); if the two columns differ, the plate has a gradient.
1% to 0.03% (0.03125%): Cells + Media containing the specified % of P80 + MTT + DMSO.

Steps

Harvest Cells

□ 1. Check confluency of Caco-2 cells under a microscope. (Ideal confluency is 70-80% in a T-150 flask.)

□ 2. Aspirate the old medium. Wash the cell monolayer once with 5 mL of sterile 1X PBS.

□ 3. Add 7 mL of 0.05 % Trypsin-EDTA and tilt the flask to cover the surface.

□ 4. Incubate at 37°C for 5 minutes. Check for cell detachment under an inverted microscope.

□ 5. Neutralize: Add 10 mL of complete culture medium (containing FBS) to the flask to inactivate the trypsin.

□ 6. Collect: Pipette the suspension up and down to break up clumps and transfer the entire volume to a 15 mL sterile conical tube.

□ 7. Centrifuge at 370 x g for 5 minutes.

□ 8. Resuspend: Aspirate the supernatant. Gently resuspend the cell pellet in 5 mL of fresh, complete DMEM. Pipette gently to create a single-cell suspension.

Prepare for Counting

□ 9. Clean the Neubauer chamber and its special coverslip with 70% ethanol and wipe dry with a lint-free wipe.

□ 10. Position the coverslip firmly over the counting grids. When pressed gently, you may see rainbow-like "Newton's rings," indicating a proper seal.

□ 11. In 96 well plate, create a 1:1 dilution of your cell suspension with Trypan Blue stain.

a. Add 10 µL of your cell suspension.
b. Add 10 µL of 0.4% Trypan Blue solution.
c. Mix gently. Your Dilution Factor (DF) is 2.

Load and count cells:

□ 12. Set a pipette to 10 µL. Place the pipette tip directly against the edge of the coverslip, just above the counting grid area. The tip should be touching both the slide surface and the side of the coverslip. The liquid will be drawn under the coverslip by capillary action. Do not overfill (liquid should not spill into the moats) or underfill.

□ 13. Let the chamber sit for 1-2 minutes to allow cells to settle onto the grid plane.

□ 14. Place the chamber on the microscope stage. Using the 10x objective, focus on the grid.

□ 15. Identify the four large 1 mm^2 corner squares (each is composed of a 4x4 grid).

□ 16. Count all the live (viable) cells (clear, bright, and refractile) in all 4 large corner squares. Ignore the dead cells, which are stained blue. To avoid double-counting, only count cells touching the top and left lines of each square. Ignore cells touching the bottom and right lines.

Calculate Cell Concentration (C1)

□ 17. Get Average: Sum the counts from the 4 squares and divide by 4.

□ 18. Average cells/square = $(Count_{Sq1} + Count_{Sq2} + Count_{Sq3} + Count_{Sq4}) / 4$

□ 19. Calculate Concentration: Use the standard Neubauer formula.

Cells/mL = Average cells/square × Dilution Factor (DF) × 10,000

Why 10,000? Each 1 mm² square has a volume of 0.1 µL, which is 1/10,000th of a mL. This factor scales your count up to cells/mL.

Example:
Counts: 50, 55, 52, 53
Average = (50+55+52+53) / 4 = 52.5
DF = 2
Concentration (C1) = 52.5 × 2 × 10,000 = **1,050,000 cells/mL**

Prepare Final Seeding Suspension and Plate

□ 20. Your goal is to seed 10,000 cells in 100 µL, which requires a final suspension (C2) of 100,000 cells/mL.

Use the $C_1V_1 = C_2V_2$ formula to calculate the volume of stock (V1) needed.

C_1 = Your stock concentration (for example, 1,050,000 cells/mL)

C_2 = Your target concentration (100,000 cells/mL)

V_2 = Final volume you need. (you will need ~50 wells x 100 µL = 5 mL).

Example:

$(1{,}050{,}000) \times V_1 = (100{,}000) \times 5$ mL

$V_1 = 500{,}000 / 1{,}050{,}000 = 0.476$ mL = 476 µL

□ 21. In a new 15 mL conical tube, add V_1 of your cell stock to (5 mL - V_1) of complete culture medium. Don't vortex. Mix well.

Cell seeding

□ 22. Add 100 µL of your final 100,000 cells/mL suspension to all designated wells (all wells except BLK and PBS wells).

□ 23. Add 100 µL of complete DMEM (no cells) to the BLK wells.

□ 24. Add 100 µL of sterile PBS to all perimeter PBS wells.

□ 25. Place the plate in the 37°C, 5% CO_2 incubator for 24 hours to allow cells to attach and resume normal metabolism.

Treatment

□ 26. After the 24-hour cell attachment, take the 96-well plate from the incubator.

□ 27. Carefully aspirate the 100 µL of seeding medium from all wells (except BLK and PBS wells).

□ 28. Using a multistep pipette, add 100 µL of your prepared P80 solutions to the appropriate wells as defined by your plate map.

□ 29. Return the plate to the 37°C, 5% CO_2 incubator for 24 hours.

Adding MTT reagent

□ 30. Remove the medium and add 100 µL fresh medium.

□ 31. Add 10 µL of MTT stock solution into each well.

□ 32. Gently swirl, wrap the plate in aluminium foil, and incubate at 37°C for 2–5 hours to allow formazan crystal formation.

Measurement

□ 33. Carefully remove 85 µL of medium from each well.

□ 34. Add 100 µL DMSO to each well.

□ 35. Pipette up and down thoroughly to dissolve crystals.

□ 36. Place on a plate shaker for 5–10 min until fully homogeneous; check visually for undissolved crystals to avoid well-to-well noise.

□ 37. Mix again by pipetting, then measure absorbance at 540 nm.

Data analysis

Blank Subtraction

□ 38. Your raw data will be the Optical Density (OD) values measured at 540 nm.

□ 39. First, calculate the average OD of your Blank (BLK) wells (which contain only media, MTT, and DMSO). This is your OD_Blank.

□ 40. Subtract this average blank value from all your other wells (Vehicle Control and P80-treated wells) to correct for the background absorbance from the media and reagents.

□ 41. Final Sample OD_{540} = Sample OD_{540} - Blank OD_{540}

□ 42. Final Vehicle Control OD_{540} = Vehicle Control OD_{540} - Blank OD_{540}

Calculate Percent Viability

□ 43. Calculate the average of your Final Vehicle Control OD_{540} values. This value

represents 100% cell viability (your "Control" group).

□ 44. Calculate the percent viability for each P80-treated sample well using the following formula:

$$\%\text{Viability} = \frac{\text{Final Sample } OD_{540}}{\text{Average Final Vehicle Control } OD_{540}} \times 100\%$$

Statistical Analysis & Visualization:

□ 45. For each P80 concentration, calculate the mean % Viability and standard deviation (SD) from your 6 replicates.

□ 46. Use statistical software (like GraphPad Prism) or RStudio to analyze the data.

□ 47. To compare each P80 concentration to the Vehicle Control (VC), use a one-way ANOVA followed by a Dunnett's post-hoc test.

□ 48. A p-value < 0.05 is typically considered statistically significant.

□ 49. Plot your data as % Viability (Y-axis) vs. P80 Concentration (X-axis). Using a logarithmic scale for the X-axis is standard for a dose-response curve and will help you visualize and calculate an IC50 value (the concentration of P80 that reduces cell viability to 50%).

References

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